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1. Some considerations are:

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- i) Cr content: controls DBTT shift, susceptibility to hardening / loss of ductility and corrosion, etc.
 - ii) Alloying constituents - A balance of the alloying constituent content is needed as some elements (Si, Ni, etc.) may provide beneficial effects (increase in ductility, strength, etc.), but too much may be detrimental.
 - precipitation, solid solution, phase stability

2. Ferritic steels swell less than austenitic steels because of the microstructural differences ^{and atomic structure}. The ferrite phase has much more free volume / GB presence which more readily accommodates FPs? thus reducing swelling. ^{void swelling!} The free volume difference arises from difference in crystal structure FCC (ferrite) vs. BCC (austenite).
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- Started down the path....

3. Ductile to brittle transition temperature (DBTT) is the temperature at which a typically ductile material becomes relatively brittle if it reaches or goes below this temperature due to the reduction in thermal diffusion processes.

- why Cr impacts this?

Irradiation embrittlement occurs at relatively low T and low dose in F-M steels. The shift of DBTT can be largely controlled through the selection of different Cr content; at a Cr content of ~ 9 wt.%, this shift is minimized, where any higher/lower content sees a parabolic increase in this shift as they stray further from 9 wt.%. Operate $> 500^\circ\text{C}$

4. FeCrAl alloys are of interest due to their corrosion resistance (formation of Al_2O_3 passivating layer + Cr) and minimal swelling (small FG bubbles).

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The main considerations for composition balance is Cr/Al content.

- Too low Cr and Al: poor corrosion resistance
- Too high Cr and Al: increased hardness/loss of ductility
- Too high Cr or Al: unfavorable precipitation formation.

5. In ODS steels, oxide particles provide the following benefits:

i) Barriers for dislocation motion - increases resistance to creep motion ✓

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ii) Sinks for FGs - FGs are attracted to the oxides which results ↓ in less FG agglomeration in fuel matrix

no fission gases!
point defects

6. Advantages

7/8 • Maintains advantageous mechanical properties at high T, including: ✓

- Strength ✓
- Creep resistance ✓
- UTS ✓

• compatible w/ coolants

Disadvantages

- He and H embrittlement from (n, α) rxns w/ Ni and Fe/Cr at fast fluences ✓
- More expensive than alternatives ✓
- Susceptible to high dose environments (not very irradiation resistant) ✓

7. Benefits

- Higher U loading ✓
- Supports lower enrichment

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Drawbacks

- Notable FG swelling ✓
- Poor high T performance ✓
- Larger fuel zone
- Lower chemical stability with surrounding Al ✓

U_3Si swelling is much more pronounced than U_3Si_2 due to the difference in free volume/viscosity that the additional Si bond introduces. The effect of the extra Si bond is reduced at higher T. $\text{viscosity} \propto \frac{1}{\text{diffusion}}$

8. The gamma phase is the dominant phase in-reactor due to the disordering of the gamma' phase (one phase formed during decomposition of gamma U-Mo). A balance between the thermodynamic tendency for decomposition and irradiation-induced defect formation is the critical point. The conditions are imposed such that the latter of that process is dominant ✓ to retain the gamma phase.

9. The solidus / liquidus gap, in the context of fabrication of U-Mo fuels, affects the formation of Mo-lean / Mo-rich regions during solidification. This is important because Mo-lean regions are more likely to decompose to alpha/gamma' phases.

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10. At low BU: minimal FG bubble presence⁺
 - Fission gas synergetics

8/10 Increasing BU: grain refinement occurs across the fuel - increased GB sinks, increased size/number^x density of FG bubbles

High BU: grain refinement completes - large FG bubbles spread throughout fuel ✓

11. The interaction layer exhibits breakaway swelling due to the formation of low density U-Mo-Al phases. To mitigate the formation of these phases, Si was added to the coating of the intermetallic fuel particles; this initially worked, but increasing T largely inhibits this strategy. Additionally, a new fuel form was explored - monolithic. This introduced a Zr interdiffusion barrier to remove interaction between fuel-Al. Zr still interacts with both materials, but the effect of breakaway swelling was largely prevented

12. Al is cheap, plentiful, easy to work with/fabricate components, excellent corrosion/hydriding resistance, is relatively neutron transparent, etc. Despite this, Al has a low elastic modulus and melting T and poor creep resistance at higher T ($>150^{\circ}\text{C}$). Considering this, the negatives outweigh the benefits in a hotter LWR environment, but this is not the case for RTRs which have much lower bulk fluid T.